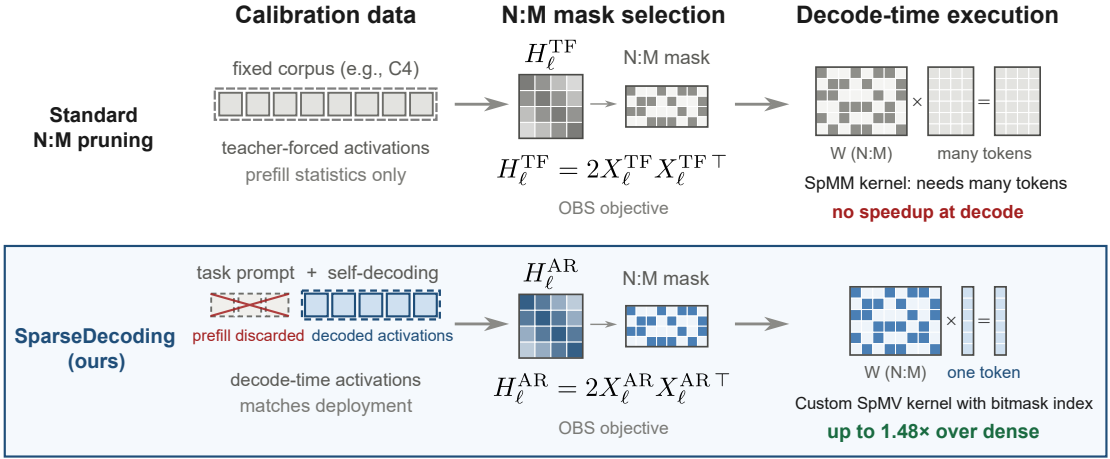

Calibration data
N:M mask selection
Decode-time execution
Standard
N:M pruning
fixed corpus (e.g., C4)
teacher-forced activations
prefill statistics only
H_ℓ^{TF}
N:M mask
$H_\ell^{\mathrm{TF}} = 2X_\ell^{\mathrm{TF}} X_\ell^{\mathrm{TF}\top}$
OBS objective
W (N:M)
many tokens
SpMM kernel: needs many tokens
no speedup at decode
SparseDecoding
(ours)
task prompt + self-decoding
prefill discarded
decoded activations
decode-time activations
matches deployment
H_ℓ^{AR}
N:M mask
$H_\ell^{\mathrm{AR}} = 2X_\ell^{\mathrm{AR}} X_\ell^{\mathrm{AR}\top}$
OBS objective
W (N:M)
one token
Custom SpMV kernel with bitmask index
up to 1.48× over dense